

Jean P. Twagirayezu, PhD

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Education

Michigan State University <i>Ph.D. in Physics — Astroparticle physics</i>	<i>Jan 2020 – May 2026</i>
Michigan State University <i>M.S. in Physics</i>	<i>Aug 2018 – Dec 2020</i>
African Institute for Mathematical Sciences <i>M.S. in Mathematical Sciences</i>	<i>Sep 2017– Jun 2018</i>
University of Rwanda, College of Education <i>Bachelor of Education (Hons.) in Physics</i>	<i>Sep 2011 – Jul 2015</i>

Research Work Experience

Graduate Research Assistant <i>Michigan State University, Pacific Ocean Neutrino Experiment</i>	<i>East Lansing, MI, USA</i> <i>Aug 2021 – Dec 2025</i>
◦ Collaborated with a team of software engineers and scientists to study the performance of the P-ONE using Monte Carlo simulations. ◦ Conducted statistical data analysis to forecast the sensitivity of the P-ONE to astrophysical point sources of neutrinos. ◦ Co-development of track event reconstruction algorithms for the P-ONE. ◦ Processed large datasets of Monte Carlo simulations from neutrino event generation to detector response simulation for the P-ONE using high-performance computing clusters. ◦ skills: statistical data analysis, software development (Python, git) · scientific computing, teamwork, and presentation. Member of the P-ONE collaboration.	
Graduate Research Assistant <i>Michigan State University, IceCube Neutrino Observatory</i>	<i>East Lansing, MI, USA</i> <i>May 2020 – Aug 2021</i>
◦ utilized the likelihood method and photon propagation for energy reconstruction of cascade events from the IceCube Monte Carlo Simulation dataset on high-performance computing clusters and GPUs. ◦ member of the IceCube Collaboration and performed one week of IceCube monitoring shifts for data quality and detector operation.	

Teaching Work Experience

Graduate Teaching Assistant <i>Michigan State University, Department of Physics and Astronomy</i>	<i>East Lansing, MI, USA</i> <i>Jan 2026 – May 2026</i>
◦ Served as a teaching assistant for PHY 222 Studio Physics for Life Scientists II. ◦ Facilitated in-class group problem solving through integrated physics laboratory and discussion, graded lab reports, and provided academic support during office hours and recitation sessions.	
Graduate Teaching Assistant <i>Michigan State University, Center for Integrative Studies in General Science</i>	<i>East Lansing, MI, USA</i> <i>Jan 2019 - May 2020</i>
◦ Taught and graded three sections of introductory physics lab, ISP 209L, supporting over 90 students per semester.	
Graduate Teaching Assistant <i>Michigan State University, Physics and Astronomy</i>	<i>East Lansing, MI, USA</i> <i>Aug 2018 – Dec 2018</i>
◦ Conducted help room sessions, assisted in class demo setup for PHY 232 Introductory Physics II.	

Physics Teacher*Kagarama Secondary School (KSS)**Kigali, Rwanda**Feb 2017 - Aug 2017*

- Taught advanced high school physics courses (electromagnetism, oscillations and waves, introduction, modern physics), lower secondary physics courses, and upper-level mathematics courses for non-majors. Designed and delivered lessons, conducted laboratory sessions, and provided assessments, improving students' performance.

Physics Teacher*Saint Paul International School (SPIS)**Kigali, Rwanda**Jan 2016 - Aug 2017*

- Taught Cambridge As & A Level Physics courses, upper-level high school physics courses (mechanics, heat and thermodynamics, geometric optics, electrostatics, electromagnetism, oscillations and waves, introduction to modern physics), and lower-level Mathematics courses. Designed and delivered course lessons, conducted physics laboratory sessions, and provided assessments, supporting high exam pass rates.

Physics Teacher*APRED Ndera Secondary School**Kigali, Rwanda**Jul 2015 - Jan 2016*

- Taught upper secondary school Physics courses, designed and delivered course lessons, conducted physics laboratory sessions, and provided assessments, enhancing curriculum delivery.

Physics Teacher*Nu-Vision High School (N.V.H.S)**Kigali, Rwanda**Sep 2014 - Jul 2015*

- Taught lower secondary school Physics classes, designed and delivered course lessons, and provided assessments, supporting student learning outcomes.

Mentoring Experience

Mentoring Undergraduate Students*Michigan State University, the P-ONE group**East Lansing, MI, USA**May 2023 - May 2025*

- Mentored two undergraduate students in the MSU P-ONE research group on projects including Monte Carlo Simulations production, neutrino event track reconstruction, and P-ONE geometry optimization studies, fostering research skills development.

Mentoring High School*UN Ivy STEM Connect - AIMS Rwanda, Lycée de Kigali**Kigali, Rwanda**Dec 2017 - Apr 2018*

- Provided mentorship to students from Lycée de Kigali, a secondary school, in Mathematics and Physics in collaboration with AIMS Rwanda students and Ivy League students studying in the USA.

Technical Skills

Languages: English (Fluent), French (Intermediate), Kinyarwanda (Native)**Programming tools:** Python, SQL, C++ (basics), shell (basics), Icecube icetray, Latex, Excel.**Data Analysis tools:** Numpy, Pandas, Scipy, Sklearn, Keras, Matplotlib, Jupyter Notebook, Git/Github**Research & Teaching:** Statistical data analysis, Machine Learning, Software development, Advanced maths, Astroparticle physics, Teaching, Communication, Presentation, Team work.